CODE: 333/67/ESS

MARKING SCHEME
SENIOR SECONDARY
MARCH-APRIL 2024
ENVIRONMENTAL SCIENCE
(333)

SET			EXPECTED VALUE POINTS FOR EACH STEP	Distribution Of Marks	Total Marks
A	B	C			
Q. No.					
			Core Modules **Objective Question Type- I**		
1	-	-	b or b	1	1
2	-	-	c	1	1
3	-	-	a or c	1	1
4	-	-	b		
5	-	-	c or b	1	1
6	-	-	a or d	1	1
7	-	-	a	1	1
8	-	-	c or a	1	1
9	-	-	b	1	1
10	-	-	a	1	1
11	-	-	c	1	1
12	-	-	b	1	1
13	-	-	a or d	1	1
			Optional Module 8A **Water Resources Management**		
14	-	-	b or a	1	1
15	-	-	b or a	1	1
16	-	-	b or a	1	1
			Optional Module 8B **Energy and Environment**		
14	-	-	c or b	1	1
15	-	-	a or b	1	1
16	-	-	a or a	1	1
			Core Modules **Objective Question Type- II**		
17	-	-	• Environmental Impact Assessment • Risk Assessment, environmental Management Post Product Monitoring (Any two) **OR** • Go ahead signal granted by the Impact Assessment Agency in MoEF, Govt. of India.	1 ½ + ½ 1	2

			• Industries, mining Any other Any two	½ + ½	
18	-	-	(1) EIA report has to be exactly in keeping with the process and procedure laid down for EIA. (2) Forests assigned to a village community by state. (3) The aim of IPM is to keep the crop damage to economically tolerable level. (4) Chemical Fertilizers and pesticides. (5) As exotic species breed extra fast due to absence of any enemy. (6) Acid rain damages cuticles of plant leaves which in turn reduce photosynthesis. (7) No life can exist without water. It needs to be compared for existence of life. (8) As they take million of years to be formed.	1×6	6
19	-	-	(1) India (2) Intellectual Property Right (3) Poverty (4) Methane (5) Arthropods (6) Ozone Hole (7) Sulphurdioxide (8) Monsoon forests	1×6	6
20	-	-	(1) Carrying capacity (2) Work (3) Biofuels (4) Ex-situ (5) Carcinogens (6) Planktons	1×4	4
21	-	-	(a) T (b) F (c) T (d) T (e) T	1×3	3
			Optional Module 8A **Water Resource Management**		
22	-	-	2.7% or approx. 3% **OR** Rivers, lakes, ground water (Any two)	1 Or ½ + ½	1
23	-	-	Storage of rainwater on surface and recharge of ground water **OR** Rooftop harvested rainwater which was directed into underground tanks.	1	1
24	-	-	Untreated household water which has not come into	1	1

<table>
<tr><td></td><td></td><td></td><td>contact with toilet waste.
OR
In Irrigation / cultivation of short term and long term crops / fish aquaculture.
(Any two)</td><td>

Or

½ + ½</td><td></td></tr>
<tr><td></td><td></td><td></td><td>Optional module 8B
Energy and Environment</td><td></td><td></td></tr>
<tr><td>22</td><td>-</td><td>-</td><td>When it will depend on cost effective/cheaper environment friendly energy forces.
OR
• Depletion of energy resources
• Pollution due to emission of given house gases.
(Any one)</td><td>1</td><td>1</td></tr>
<tr><td>23</td><td>-</td><td>-</td><td>Left over of sugar cane after it is crushed and its juice extracted.
OR
A series of chemical reactions in presence of methanogenic bacteria to produce CH_4 and CO_2.</td><td>1</td><td>1</td></tr>
<tr><td>24</td><td>-</td><td>-</td><td>Uneven absorption of solar radiation of earth's surface causes difference of temperature ; density of pressure which turns air into wind.
Or
It is the solar energy of sun which causes kinetic energy of wind</td><td>½+½

Or

1</td><td></td></tr>
<tr><td></td><td></td><td></td><td>Core Modules
Subjective Part</td><td></td><td></td></tr>
<tr><td>25</td><td>-</td><td>-</td><td>(i) Assessment of existing environmental status
(ii) Various factors of eco system
(iii) Analysis of adverse environmental impact of the proposed project
(iv) Impact on people in the neighbourhood.</td><td>½ × 4</td><td>2</td></tr>
<tr><td>26</td><td>-</td><td>-</td><td>The rate at which the trees in a forest are being cut for; exceeds the rate at which trees grow. So forests are getting denuded.
OR
Because the natural habitats of wild life are being destroyed; due to deforestation.</td><td>1+1</td><td>2</td></tr>
<tr><td>27</td><td>-</td><td>-</td><td><table><tr><th>Factor</th><th>Rural</th><th>Urban</th></tr><tr><td>Occupation</td><td>Agriculture</td><td>Industry, traders professionals</td></tr><tr><td>Population Density</td><td>Comparatively low</td><td>Large</td></tr></table></td><td>1x2</td><td>2</td></tr>
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28	-	-	• The desert, • Plant-Khejri • Animal – Indian gazelle, • Bird-great Indian bustard (Any other correct examples for all)	½ × 4	2
29	-	-	- to prevent the loss of genetic diversity of species - to save a species from becoming extinct - to protect ecosystems damage and degradation Or Any other (Any two)	1x2	2
30	-	-	- Reduction in loss and wastage of water - Harvesting, collecting and improved storage of water	1+1	2
31	-	-	Mutualism – close association between two species in which both the species benefit. Commercialism – In this relationship one species benefit while other is neither harmed nor benefitted. **OR** Process by which communities of plants or animal species in an area are replaced or changed into another over a period of time. Primary and secondary succession.	1 1 Or 1 ½+ ½	2
32	-	-	(i) Rhizobium – It is a symbiotic bacteria fixes nitrogen and releases surplus fixed nitrogen to fertilize the soil. (ii) Azospirillum – fixes nitrogen and releases growth hormones and vitamins. **OR** (i) Development of high yielding varieties of crops. (ii) Crops for better nutritional quality (iii) Crops resistant to diseases and pests (iv) Improving varieties against heat, cold, frost, drought etc. (Any two)	1+1	2
			Core Modules **Long Answer Type - I**		
33	-	-	• Group of individuals living in the same given area and capable of interbreeding and sharing genetic material. • Importance to provide data on total manpower available -To estimate the total amount of foods/services required -To promote cultural / regional/ linguistic harmony	1 ½ ½	4

			(Any one) • **Major periods:** -An early period of hunters and gatherers. -Period of rise of agriculture -Period of industrial revolution -Present time **OR** Urbanization leads to loss of environment pollution of air, water and soil (i) Forests - forests are cut their area shrink (ii) Socio economic factors – population is redistributed / unbalanced urban social evils (iii) Availability of water – leads to scarcity.	½ X4 Or 1 1x3	
34	-	-	Parts of natural and cultural landscapes extending over large areas which are internationally recognized within UNESCO's man and biosphere program. Three zones– core, buffer, transition **Function**– • Conservation of ecosystems development, • Promote sustainable economic scientific research, • Monitoring and education on conservation. (Any two)	1 1 1x2	4
			Core Modules **Long Answer Type - II**		
35	-	-	The ratio between amount of energy acquired from the lower trophic level and the amount of energy transferred from higher trophic level is ecological efficiency. At each trophic level a part of available energy is lost in respiration and metabolism. A part of energy is lost as heat Herbivores – 10 Cal Carnivores – 1 Cal	2 1 1X2	5
36	-	-	Sound without value / Any noise that is unnoted by the recipient. Indoor sources and outdoor sources. **Effects of noise pollution:** Disturbs once of sleep, annoyance and treatability.	1 ½ + ½ 1X2	5

			Causes hypertension, emotional problems, efficiency and performance decreases (Any two) **Ways** • Proper maintenance of vehicles • Planting a green body trees Or any other Any two ways **OR** (i) Pollutants are discharged from a specific location directly into a water body. (ii) Discharge of hot water from power plants into rivers to increase the temperature. (iii) Phenomenon of nutrient enrichment of a water body. (iv) Waste water from domestic or industrial source or from garbage dumps. (v) Discharge of pollutants from diffused source or from a larger area like fields roads etc.	½ X2 **Or** 1 1 1 1 1	
			Optional Module 8A **Water Resource Management**		
37	-	-	• Domestic water supply • Irrigation • Industrial use (Any two)	1x2	2
38	-	-	• Droughts • Irrigation demand • Industrial demand • Pollution • Waste (Any four) **OR** Decreased community dependency on ground water; reduction of possibility of floods; suited for landscape irrigation; conservation of groundwater reduction of drainage problem. Or Any other. Any four	½ × 4	2
39	-	-	Watershed that contribute water to a stream or water body An area through run off or underground path Water moves through soil and layers which act as filters. OR Process by which ground water reservoir is augmented at a rate exceeding normal rate of seepage	1 1 1	2

			and replenishment **Methods** • Water spreading • Recharge through pits • Rooftop collection • Road top collection • Induced recharge (Any two methods)	½ X2	
			Optional Module 8B **Energy and Environment**		
37	-	-	Device that converts solar energy into electrical energy. • Used for domestic lighting • Street lighting • Village lighting • Water pumping • Electrification(Any two)	1 ½ + ½	2
38	-	-	Natural heat from interior of earth that can be converted to generate electricity and heat up buildings. • Volcanoes • Hot springs • Geysers (Any two) **OR** These are electrochemical devise that convert the chemical energy of a fuel directly into electricity; So it reduces urban air pollution.	1 ½ + ½ Or 1x2	2
39	-	-	It is a systematic approach to monitor industrial energy consumption and to find out sources of energy wastage. To identify the areas of energy consumption; find the overuse for accessing the opportunity of saving energy; to make the need for energy conservation. **OR** A township oriented towards ecofriendly and energy conserving life. Can be created by creating awareness for energy conservation; by developing right attitude in residents; and by grants on infrastructures.	1 ½ X 2	